

A Concentration-Based Drift Calibration Transfer Learning Method for Gas Sensor Array Data

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Outline

Introduction

Proposed Method

Experimental Results

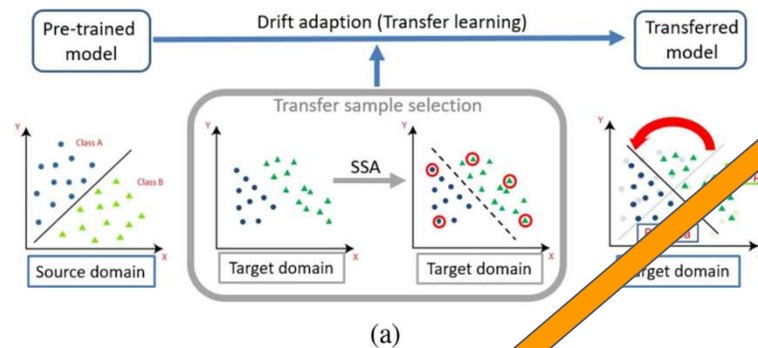
Validation on External Dataset

Conclusion

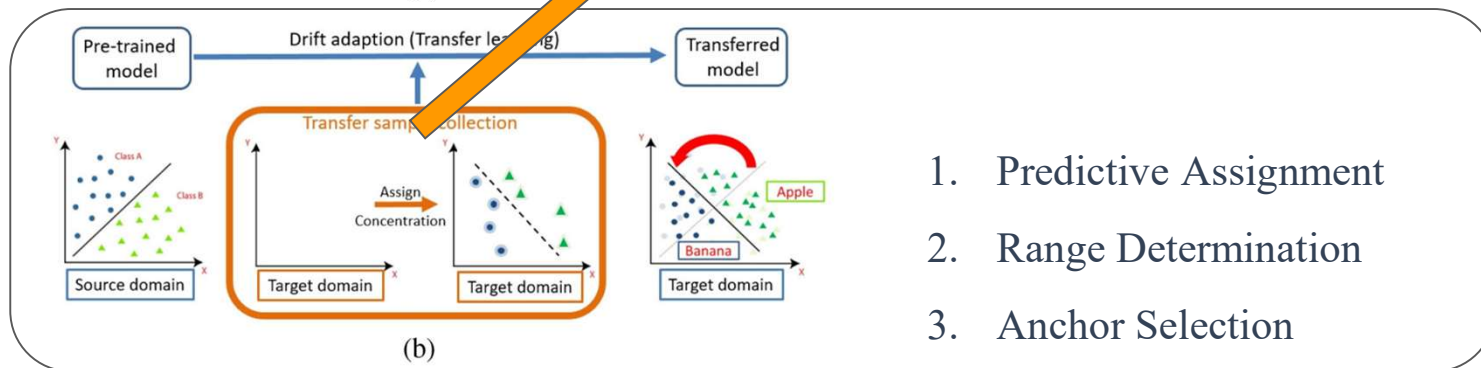
Introduction

- The Drift Challenge
 - Sensor aging and environmental factors cause data distribution shifts
 - Classification accuracy decreases over time
- Proposed Framework: CDC
 - Utilizing gas concentration as an anchor.
 - Unlike traditional methods that adjust decision boundaries, CDC recalibrates the underlying scale of the sensor response.
- Application
 - Operates continuously without prior knowledge of future data distributions.
 - Ensures long-term stability under severe sensor decay.

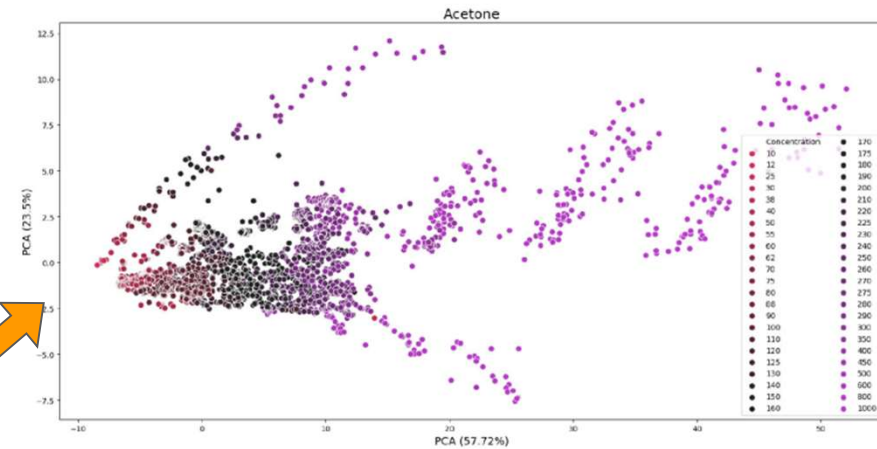
Selection Strategy



(a)



(b)

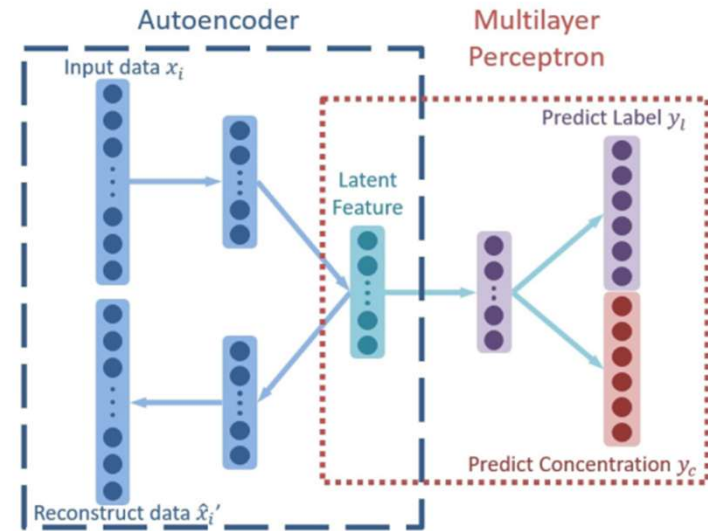


Gases with higher concentrations consistently distribute towards the right side of the PCA plot, even in late-stage batches.

1. Predictive Assignment
2. Range Determination
3. Anchor Selection

CB Model

- **Architecture**
 - Autoencoder(feature extractor)
 - 64x30x20x30x64
 - MLP(predictive head)
- **Multi-Task Ground Truth**
 - label & concentration(12D)
- **Loss function**
 - **Reconstruction**
 - **Identification**
 - **Physical Scaling**



$$J_{\text{all}} = J_{\text{AE}} + J_{\text{class}} + J_{\text{con}}$$

$$= \sum_i (\|x_i - \hat{x}_i\| + \text{BCE}(t_l, y_l) + \|t_c - y_c\|)$$

$$t_{l,j} = \begin{cases} 1, & j = g \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$t_{c,j} = \begin{cases} 0.01 \times c, & j = g \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$t = (t_l, t_c) \quad (3)$$

CB Drift Calibration Transfer Process

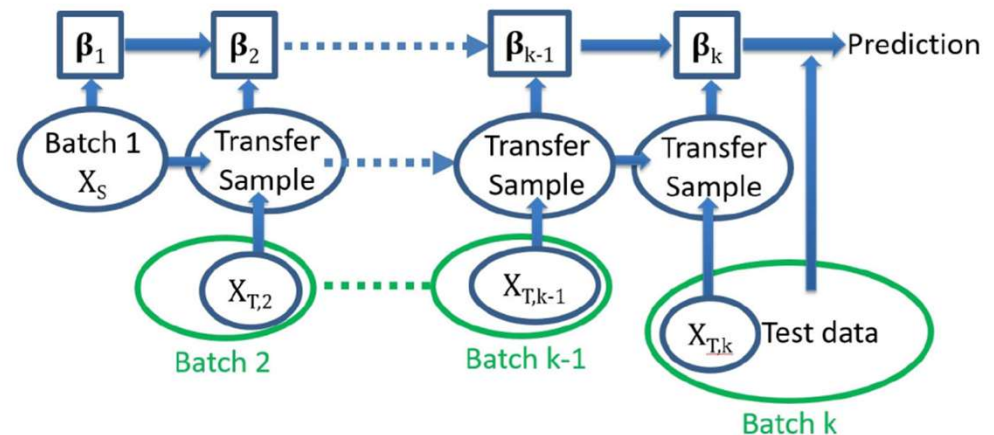
1. Data Retention

- a. cumulative learning
- b. expanding pool

2. Time-Decay Weights

3. Recursive Model Adaption

- a. Sequential update
- b. Online adaption



Experimental Results- CB Models

TABLE 1. Comparison of Transfer Accuracies of Different Models

Batch ID	2	3	4	5	6	7	8	9	10	Avg.
No transfer	85.77	67.40	44.10	34.01	44.74	35.59	15.99	33.83	35.47	44.1
CC-PCA [11]	90.92	40.86	47.20	59.39	56.74	56.71	36.39	45.32	37.72	52.36
OSC [12]	88.10	66.71	54.66	53.81	65.13	63.71	36.05	40.21	40.08	56.50
ensemble [15]	74.36	87.83	93.79	95.43	69.17	69.72	91.84	76.38	65.50	80.45
ML-comGFK [19]	80.25	74.99	78.79	67.41	77.82	71.68	49.96	50.79	53.79	63.28
DRCA [13]	89.15	92.69	87.58	95.94	86.52	60.25	62.24	72.34	52.00	77.63
DAELM-S(30) [16]	87.98	95.74	85.16	95.99	99.14	83.51	86.90	100	53.62	87.00
TCTL [17]	96.95	96.85	91.30	98.98	86.78	82.51	86.05	83.19	65.75	87.60
DCAE-CL [14]	97.47	96.13	90.50	98.58	94.94	95.47	92.62	96.40	76.82	93.21
CB model*	95.82	97.86	93.79	96.95	90.70	87.35	92.52	93.19	78.50	91.85

*CB model with five transfer data of each class. ** Bold values indicate the best result of each batch.

Experimental Results- CDC Transfer Process

Batch ID	Overall accuracy (%)								
	2	3	4	5	6	7	8	9	10
Transfer setting 1	60.45	58.49	47.83	50.14	55.67	66.76	25.20	40.68	71.13
CDC transfer process	57.51	52.33	60.20	61.62	58.17	72.99	69.91	78.53	75.61

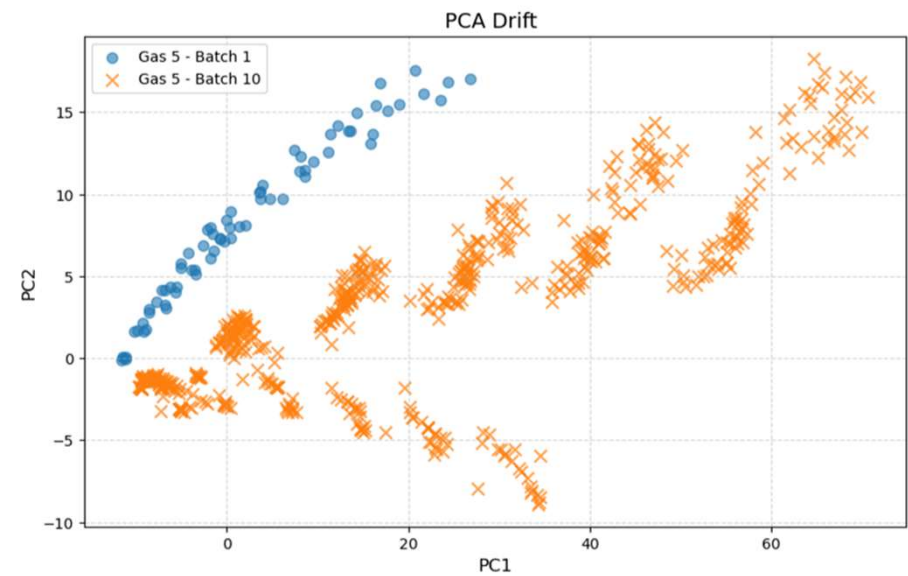
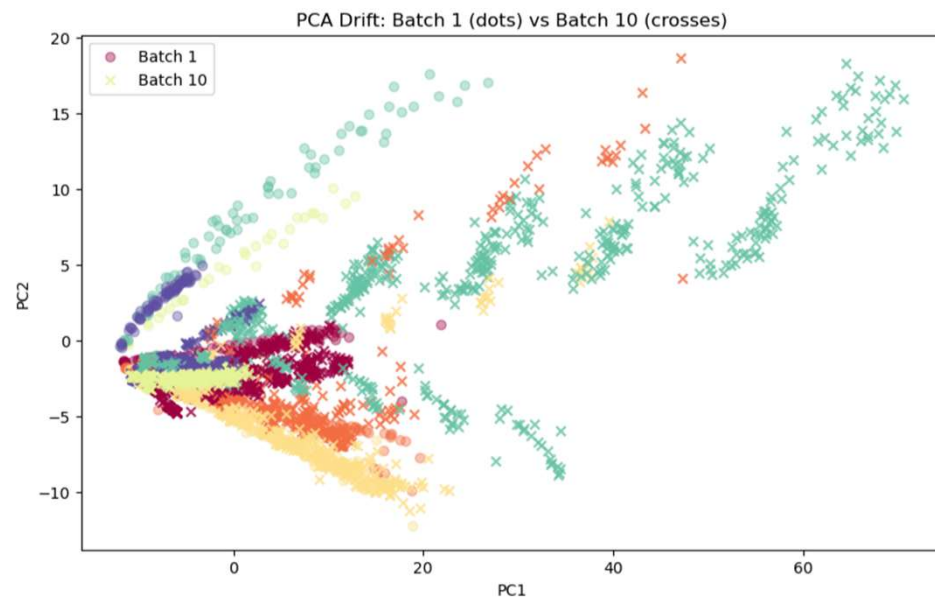
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- **Transfer Setting 1: Resets to Batch 1 weights every time and fine-tunes only on the latest batch, ignoring the historical drift trajectory.**
- **CDC Transfer: Inherits weights from the previous state and uses a cumulative, time-weighted sample pool to maintain continuous adaptation.**

Validation- UCI Gas Sensor Drift Dataset

- Time Span: 36 months (Jan 2007 – Feb 2011).
- Sensors: 16 chemical sensors (metal-oxide)
- Gas Species: 6 pure gases — Ammonia, Acetone, Acetaldehyde, Ethylene, Ethanol, and Toluene.
- Concentration Range: Wide dynamic range from 5 to 1000 ppmv.

Validation- UCI Gas Sensor Drift



Concentration gradient direction remains invariant despite signal loss.

Experimental Results- CB Models

Batch ID	2	3	4	5	6	7	8	9	10	Avg.
CB model	91.88	98.87	96.89	97.97	98.09	97.56	95.58	97.02	81.08	91.55

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Validation- UCI Gas Sensor Drift

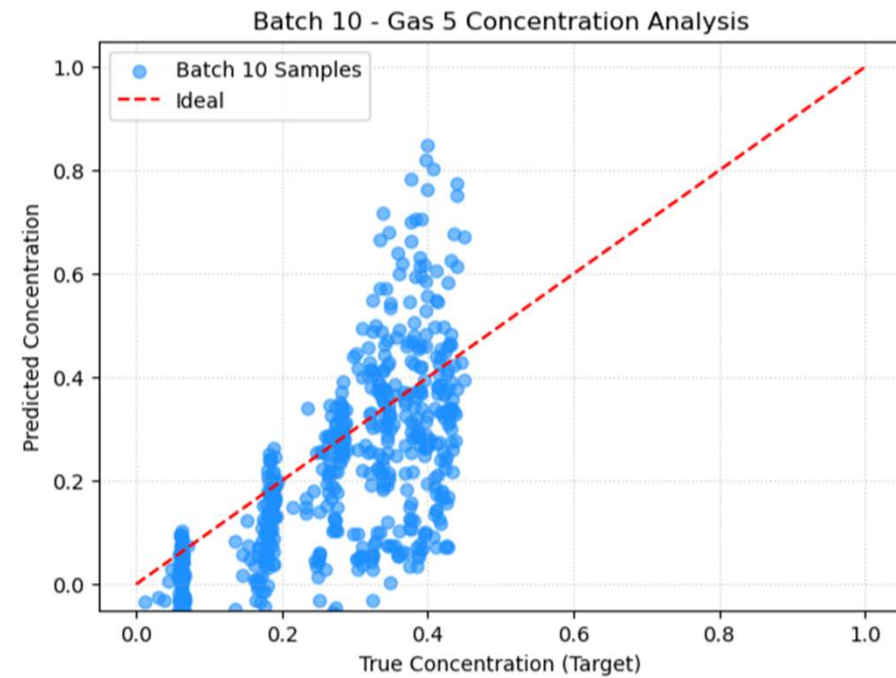
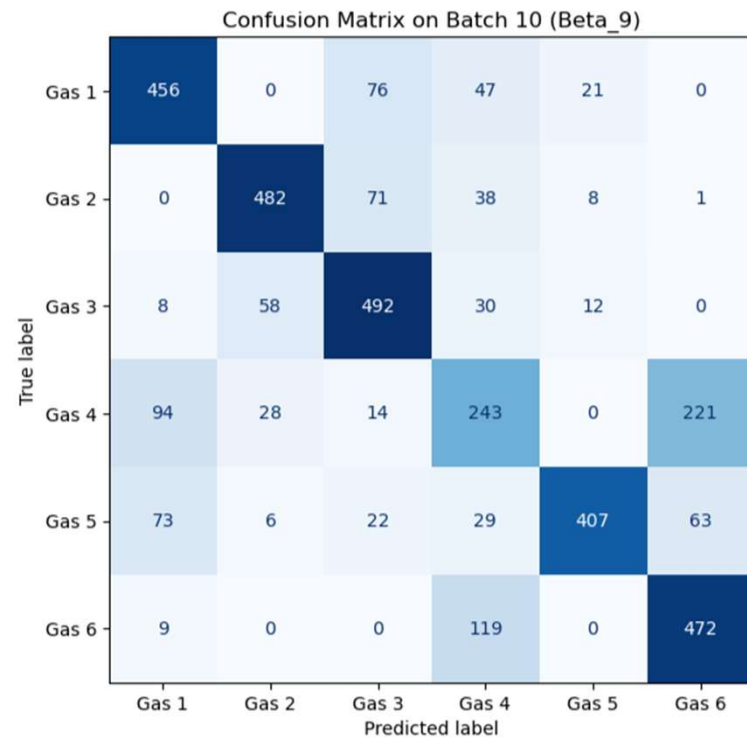
Overall accuracy									
Method / Batch ID	2	3	4	5	6	7	8	9	10
Transfer Setting 1 (%)	47.02	48.17	46.87	44.18	60.63	71.80	56.97	52.60	60.75
CDC Transfer Process (%)	43.18	58.01	56.79	60.60	78.46	83.07	87.50	87.96	92.54

- Static Fine-tuning(setting1): High instability due to loss of historical drift trajectory.
- CDC Framework: Progressive adaptation reaching peak accuracy at Batch 10.

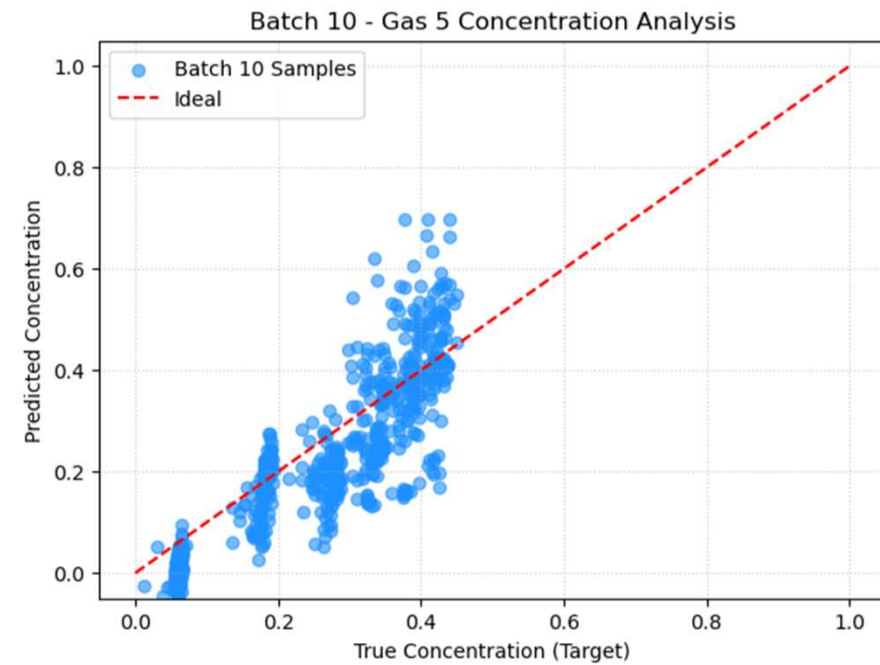
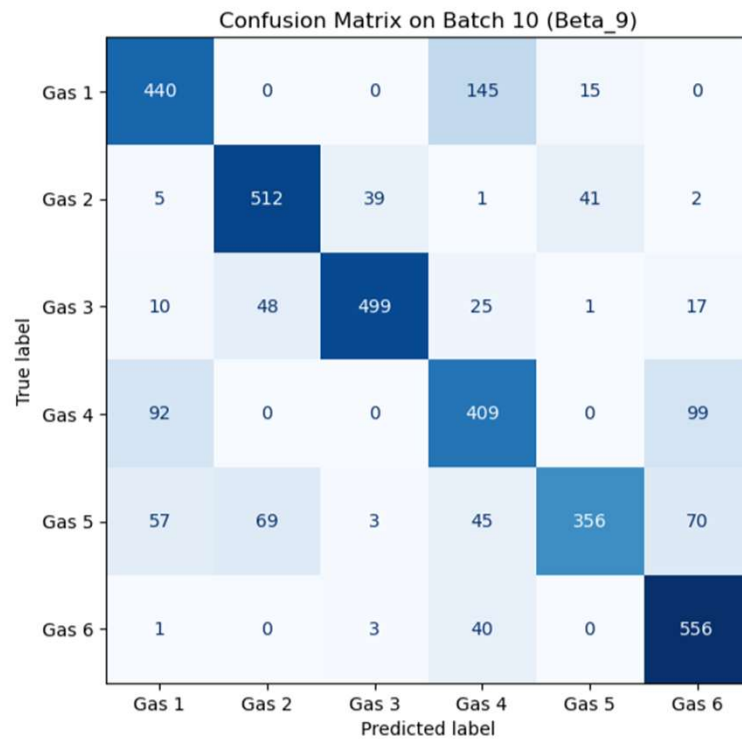
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Validation- UCI Gas Sensor Drift (setting1)



Validation- UCI Gas Sensor Drift (CDC Transfer)



Batch 10 Accuracy: 0.9817

Conclusion| Future Work

- The Power of Anchors: High-concentration samples serve as reliable Physical Baselines for real-time encoder calibration.
- Zero Future Knowledge: The framework operates strictly in a recursive manner, making it deployable for real-time, long-term monitoring without needing to see future data.
- Superior Accuracy: Achieved a total average accuracy of 93% and an overall transfer accuracy of 72.01%, significantly outperforming traditional static fine-tuning.
- Automated Parameter Tuning
- Analysis on Cross-Sensor Arrays